Propeller Autofeather System (Mod 6/2022) – General Data

1. General

Automatic propeller system is provided which automatically feathers the propeller of an underpowered engine when a decrease in torque to below approximately 20 psi is detected. Subsequent autofeathering of the other propeller is prevented by a blocking relay which disarms the autofeather system after the first propeller has feathered. The system is armed for operation when the following three conditions have been met:

1 AUTO FEATHER switch is selected ON

2 Torque has risen above a specific threshold on both engines, and;

3 The power levers are advanced beyond 86 – 88% N_G

Pulling either or both back below 86 – 88% N_G (for example, in the event of a rejected take-off) disarms the autofeather system. Two autofeather indicator lights illuminate to signify when the system is selected on and when it is armed. An autofeather test switch is provided to permit a ground check of the autofeather system. The autofeather system is powered from the left DC bus and is protected by a PROPELLER AUTOFEATHER circuit breaker on the main circuit breaker panel.

A two-second delay between the detection of low torque and automatic feathering of the propeller is provided. This allows propeller discing to be used in the event of a rejected take-off due to a loss of torque. Dual circuitry is used on all autofeather sensors and relays to reduce the probability of a single system failure causing an unwanted autofeathering.

2. Description

The system comprises of two propeller autofeather push button switch, autofeather test switch, indication lights, two power lever-operated microswitches, two arming light relays, two bleed resistors, two time delay relays, and a torque pressure switch and feathering solenoid valve installed on each engine. The system obtains power through a 5–ampere circuit breaker from the left main DC bus.

A. Propeller Autofeather Switch

Two identical propeller autofeather switches are provided, located outboard of each pilot's primary flight display. When activated (accomplished by pressing either switch, selected autofeather armed) electrical power is supplied to the autofeather system as indicated by the illumination of SELECT on two switches. The dual switch design eliminates the need for the right seat pilot to stretch across the instrument panel to operate the switch.

The two switches operate in the same manner as a household three-way switch system that is found at the top and bottom of a flight of stairs. Pressing either switch once changes the state of the system, from ON to OFF or from OFF to ON as appropriate.

B. Autofeather Test Switch

The propeller autofeather test switch is located on the left sub panel and is a guarded two-position momentary-on switch labelled AUTOFEATHER. The switch is spring loaded to the center off position and has an unmarked test (up) position. The test switch permits a ground test of the autofeather system with engines operating at a low speed, by bypassing the power lever-operated microswitches.

C. Power Lever-Operated Microswitches

Two microswitches are installed on a slide assembly located in the cabin roof. A slide, attached to each engine power lever cable, operates the appropriate microswitch when the power levers are advanced. Electrically both microswitches are connected in series.

D. Arming Light Relay

Two arming light relays are installed in an autofeather relay box in the cabin roof. When both relays are energized, by operation of the left and right torque pressure switches, they connect electrical power to the SELECTED AUTOFEATHER ARMED lights on the pilot's flight instrument panel.

E. Time Delay Relay

Two time delay relays, one for each engine, are installed in the autofeather relay box in the cabin roof. When energized, the time delay relay, after a time delay of two seconds, connects electrical power to its respective feather solenoid valve.

F. Feathering Solenoid Valve

A solenoid-operated valve is installed on each engine's overspeed governor. When operated electrically, the valve opens to allow oil pressure, acting on the pitch change mechanism of the propeller installed on that engine, to be relieved, so causing the propeller to feather.

G. Torque Pressure Switch

A double pole, low pressure torque switch is connected to the engine torque transmitter on each engine. The switch completes an electrical circuit to its respective time delay relay, if the torque pressure of its engine falls below 11 to 13 psi. The switch also controls its respective arming light relay when it operates on increasing torque pressure at 17 psi.

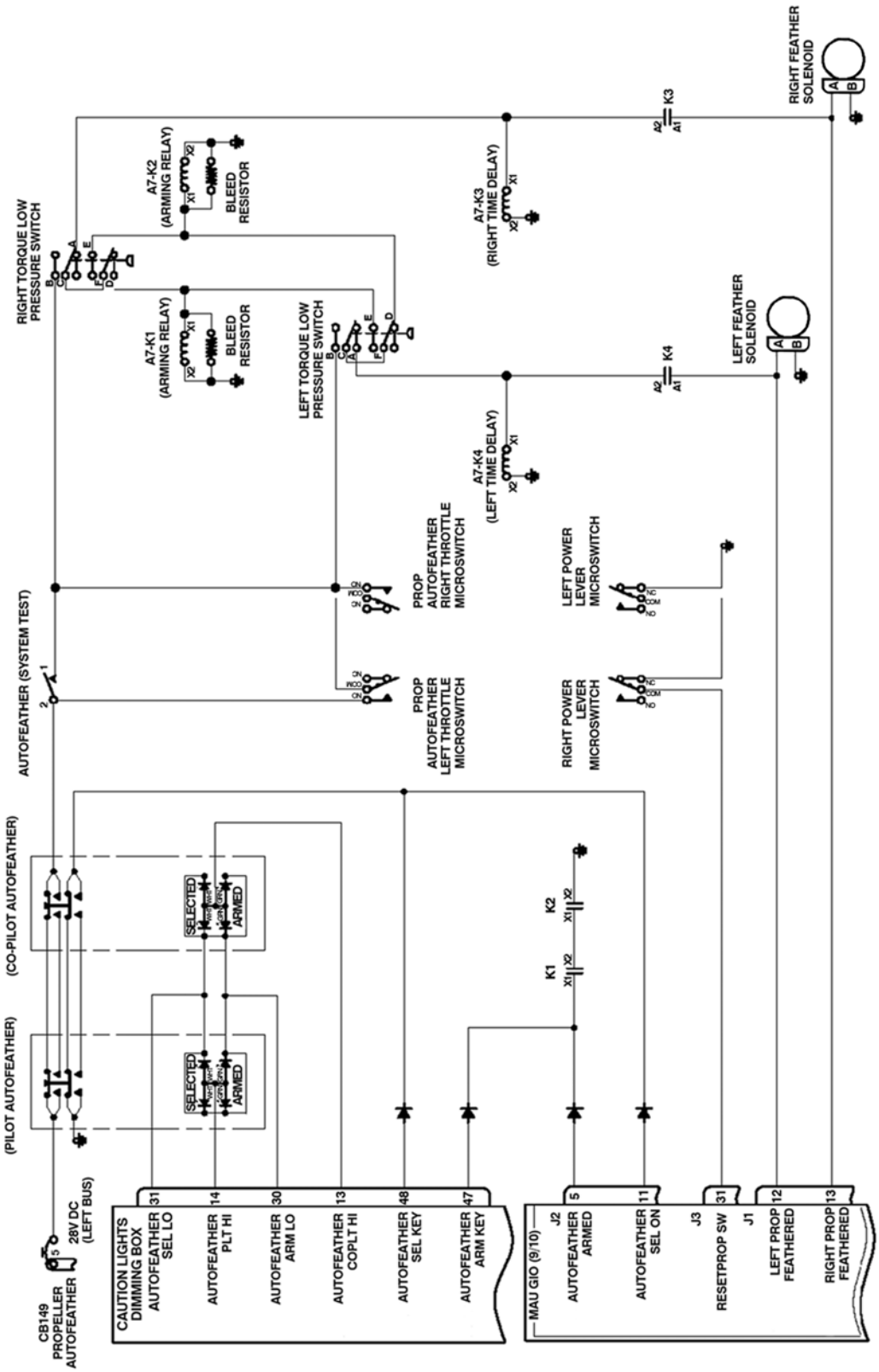
3. Operation

Refer to Figure 1 for details.

The propeller autofeather indicator lights are integral with the autofeather switches. The indicators consist of two rectangular lights, one above the other, in each switch; the upper light is labelled SELECTED and when illuminated indicates that the system has been selected on. The lower light is labelled ARMED and when illuminated indicates that the autofeather system is armed for operation. The ARMED indicator light illuminates when the autofeather is on and both power levers are advanced to 86 – 88% N_G or beyond; it goes out if either or both power levers are retarded and/or when automatic feathering of a propeller has occurred. The autofeather indicator lights are included in the caution lights test and intensity switch circuit. To comply with contemporary guidelines for the use of colour in switches and indicators, the SELECTED light is white and the ARMED light is green.

With the PROPELLER AUTOFEATHER circuit breaker engaged and the SELECTED AUTOFEATHER ARMED switch selected to ON, the selected white SELECTED indicator light illuminates. If both engine power levers are advanced to obtain 86 to 88% gas generator rpm (N_G), the two power lever-operated microswitches operate and electrical supply is provided to both torque pressure switches. Increasing torque pressure (by 17 psi) in each engine operates the torque pressure switches to energize the arming light relays K1 (by right torque pressure switch) and K2 (by left torque pressure switch) which cause the ARMED indicator green light to illuminate. In the event of the torque pressure of one engine falling to below 11 to 13 psi, when both power levers are set above 86% N_G, the torque pressure switch of that engine closes to complete a circuit to its respective time delay relay (K4 for the left engine or K3 for the right engine), which after a delay of approximately two seconds, energizes the feather solenoid valve of that engine resulting in the propeller being feathered. The closing of the torque switch also de-energizes one of the arming light relays and the ARMED indicator light goes out.

If with one propeller feathered, the torque pressure of the opposite engine should fall below 11 to 13 psi, the torque pressure switch of that engine will close and disarm the autofeather system. The feather solenoid valve of the feathered propeller is de-energized and the propeller will unfeather.



Propeller Autofeather – Electrical Schematic
Figure 1

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4. Adjustment/Test

   Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data, Para 10.B.12) for Adjustment/Test of the Autofeather System.